

7.2 v.p. V polyn. st. ≤ 2 zvolme bázi $C = (1, x, x^2)$
 $\Phi(p) = 2p' + x^2 p''$ $[\Phi]_B^B = J = ?$

$$\Phi(1) = 2(0) + x^2(0) = 0$$

$$\Phi(x) = 2 \cdot 1 + x^2(0) = 2$$

$$\Phi(x^2) = 2(2x) + x^2 \cdot 2 = 2x^2 + 4x$$

$$[\Phi]_C^C = A = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\text{rank } A = 2$$

linearita derivaci

spočítáme vlastní čísla:

$$\det(A - \lambda I) = (-\lambda)(-\lambda)(2-\lambda) = -\lambda^2(2-\lambda)$$

$$\lambda_1 = 0 \text{ (alg. násobnost 2)}, \lambda_2 = 2 \text{ (alg. násobnost 1)}$$

Pro $\lambda = 0$:

$$\dim(\text{Ker } A) = 3 - \text{rank}(A) = 3 - 2 = 1$$

$\Rightarrow$ 1 báze vel. 2

Pro $\lambda = 2$:

alg. nás. 1 $\Rightarrow$ 1 báze vel. 1

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

spočítáme vlastní vektory:

$$(A - 0 \cdot I) \vec{x} = \vec{0}$$

$$\begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_2 \\ 4x_3 \\ 2x_3 \end{pmatrix} = \vec{0} \quad x_2 = 0, x_3 = 0 \quad \vec{x} = \begin{pmatrix} t \\ 0 \\ 0 \end{pmatrix}$$

$$(A - 0 \cdot I) \vec{y} = \vec{x}$$

$$\begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2y_2 \\ 4y_3 \\ 2y_3 \end{pmatrix} = \begin{pmatrix} t \\ 0 \\ 0 \end{pmatrix} \quad y_3 = 0, 2y_2 = t, y_1 = s \quad \vec{y} = \begin{pmatrix} s \\ \frac{t}{2} \\ 0 \end{pmatrix}$$

$$(A - 2I) \vec{z} = \vec{0}$$

$$\begin{pmatrix} -2 & 2 & 0 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} -2z_1 + 2z_2 \\ -2z_2 + 4z_3 \\ 0 \end{pmatrix} = \vec{0} \quad \begin{matrix} z_1 = 2r \\ z_2 = 2r \\ z_3 = r \end{matrix} \quad \vec{z} = \begin{pmatrix} 2r \\ 2r \\ r \end{pmatrix}$$

zvolme $t=2, s=0$ a $r=1$:

$$B = (2, x, 2+2x+x^2)$$